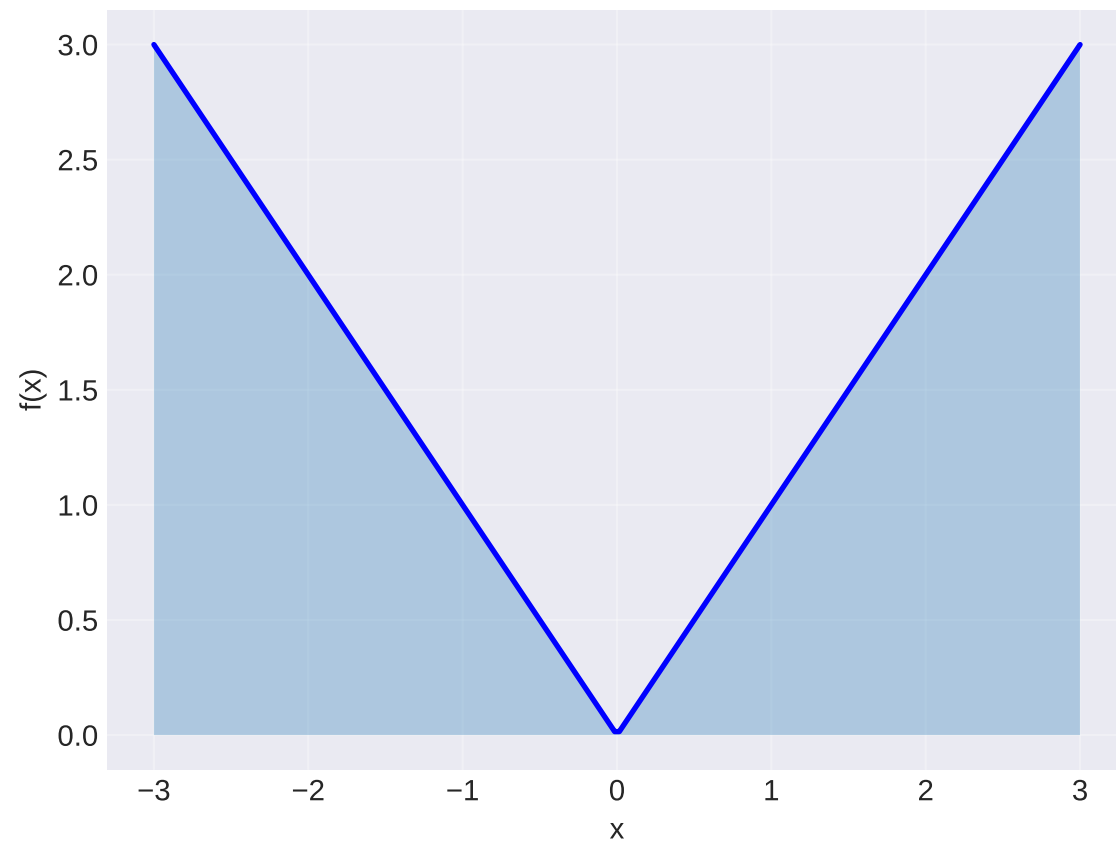
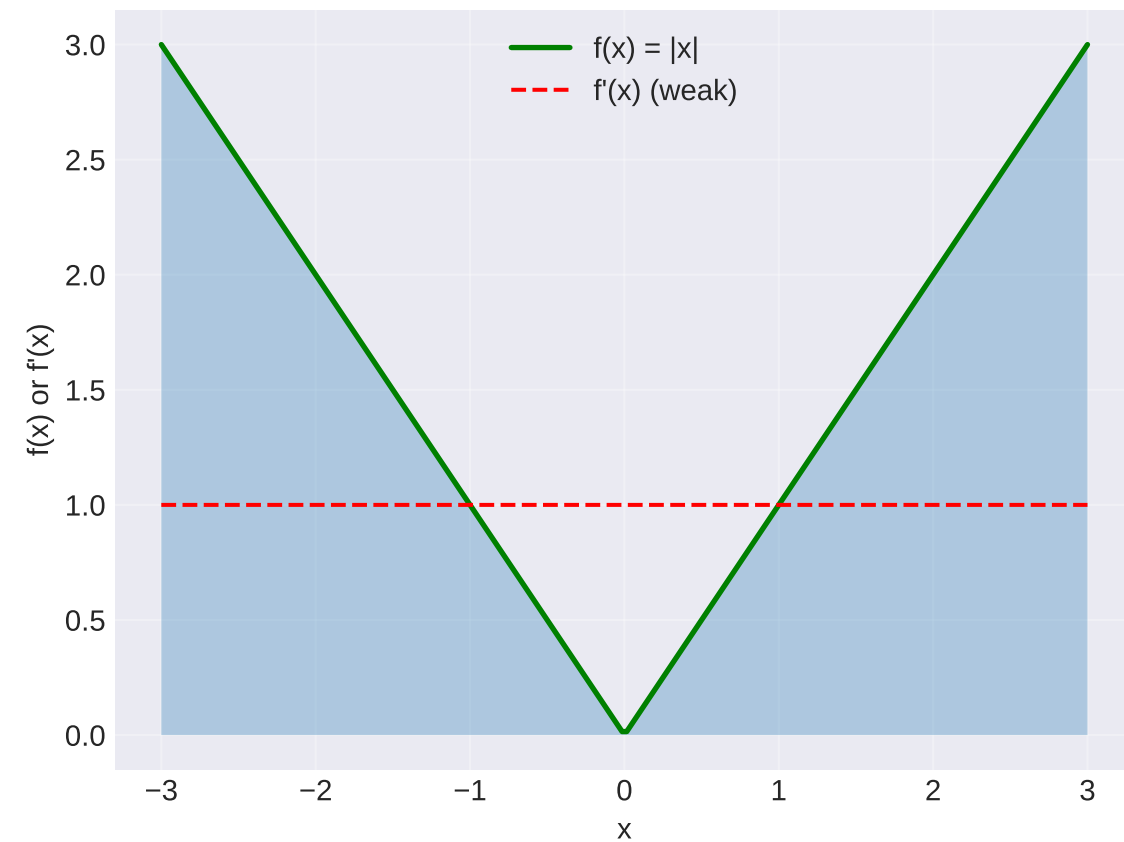


Sobolev Spaces: Functions with Weak Derivatives

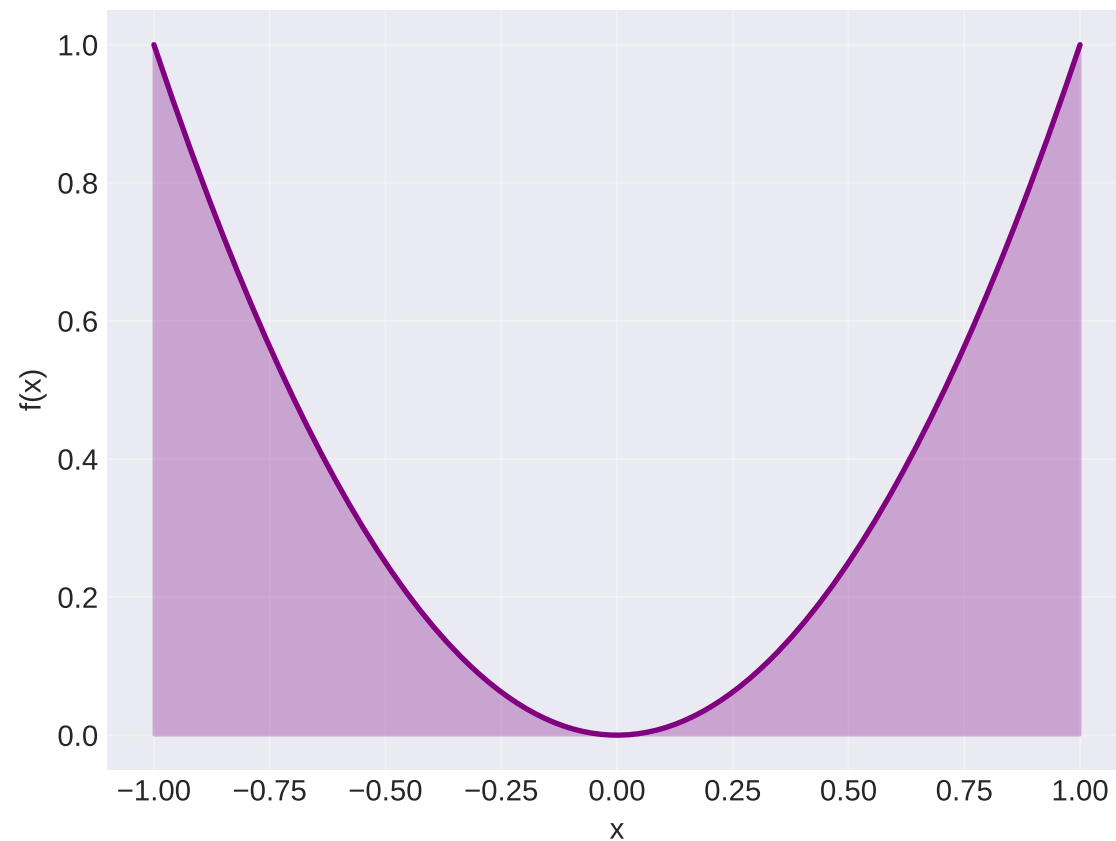
$$W^{0,p}(\Omega) = L^p(\Omega)$$



$$W^{1,p}(\Omega): \text{Functions with } L^p \text{ weak derivatives}$$



$$W^{1,p}(\Omega) \subset C(\Omega) \text{ under conditions}$$



Sobolev Space $W^{k,p}(\Omega)$:

Definition:

$$W^{k,p}(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for all } |\alpha| \leq k\}$$

Norm:

$$\|u\|_{k,p} = \left(\sum_{|\alpha| \leq k} \int_{\Omega} |D^\alpha u|^p dx \right)^{1/p}$$

Properties:

- Complete normed space (Banach space)
- Hilbert space when $p=2$ (H^k)
- Sobolev embedding theorem relates regularity to integrability